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Demonstration of summation

$$\boxed{\sum_{k=r}^{\infty} \frac{1}{\binom{k}{r}} = \frac{r}{(r-1)}}$$

Proof:

$$\text{Beta}(a, b) = \int_0^1 x^{(a-1)} \cdot (1-x)^{(b-1)} dx$$

If we choose $a = r, b = k - r + 1$

$\Rightarrow$

$$\text{Beta}(r, k - r + 1) = \int_0^1 x^{(r-1)} \cdot (1-x)^{(k-r)} dx$$

We can use other definition of Beta function:

$$\text{Beta}(r, k - r + 1) = \frac{\Gamma(r) \cdot \Gamma(k - r + 1)}{\Gamma(k + 1)} = \frac{(r-1)! \cdot (k-r)!}{k!} = \frac{1}{r} \cdot \frac{r! \cdot (k-r)!}{k!} = \frac{1}{r} \cdot \frac{1}{\binom{k}{r}}$$

Using both equations

$\Rightarrow$

$$\int_0^1 x^{(r-1)} \cdot (1-x)^{(k-r)} dx = \frac{1}{r} \cdot \frac{1}{\binom{k}{r}}$$

$\Rightarrow$

$$\frac{1}{\binom{k}{r}} = r \cdot \int_0^1 x^{(r-1)} \cdot (1-x)^{(k-r)} dx$$

$\Rightarrow$

$$\begin{aligned}
\sum_{k=r}^{\infty} \frac{1}{\binom{k}{r}} &= \sum_{k=r}^{\infty} \left(r \cdot \int_0^1 x^{(r-1)} \cdot (1-x)^{(k-r)} dx \right) = r \cdot \int_0^1 \frac{x^{(r-1)}}{(1-x)^r} \cdot \left[\sum_{k=r}^{\infty} (1-x)^k \right] dx \\
&= r \cdot \int_0^1 \frac{x^{(r-1)}}{(1-x)^r} \cdot \left[\sum_{k=0}^{\infty} (1-x)^{k+r} \right] dx = r \cdot \int_0^1 \frac{x^{(r-1)}}{(1-x)^r} \cdot (1-x)^r \left[\sum_{k=0}^{\infty} (1-x)^k \right] dx \\
&= r \cdot \int_0^1 x^{(r-1)} \cdot \left[\sum_{k=0}^{\infty} (1-x)^k \right] dx = r \cdot \int_0^1 x^{(r-1)} \cdot \frac{1}{x} dx = r \cdot \int_0^1 x^{(r-2)} dx \\
&= r \cdot \left[\frac{x^{(r-1)}}{(r-1)} \right]_{x=0}^{x=1} = \frac{r}{(r-1)}
\end{aligned}$$

$\Rightarrow$

$$\sum_{k=r}^{\infty} \frac{1}{\binom{k}{r}} = \frac{r}{(r-1)}$$
